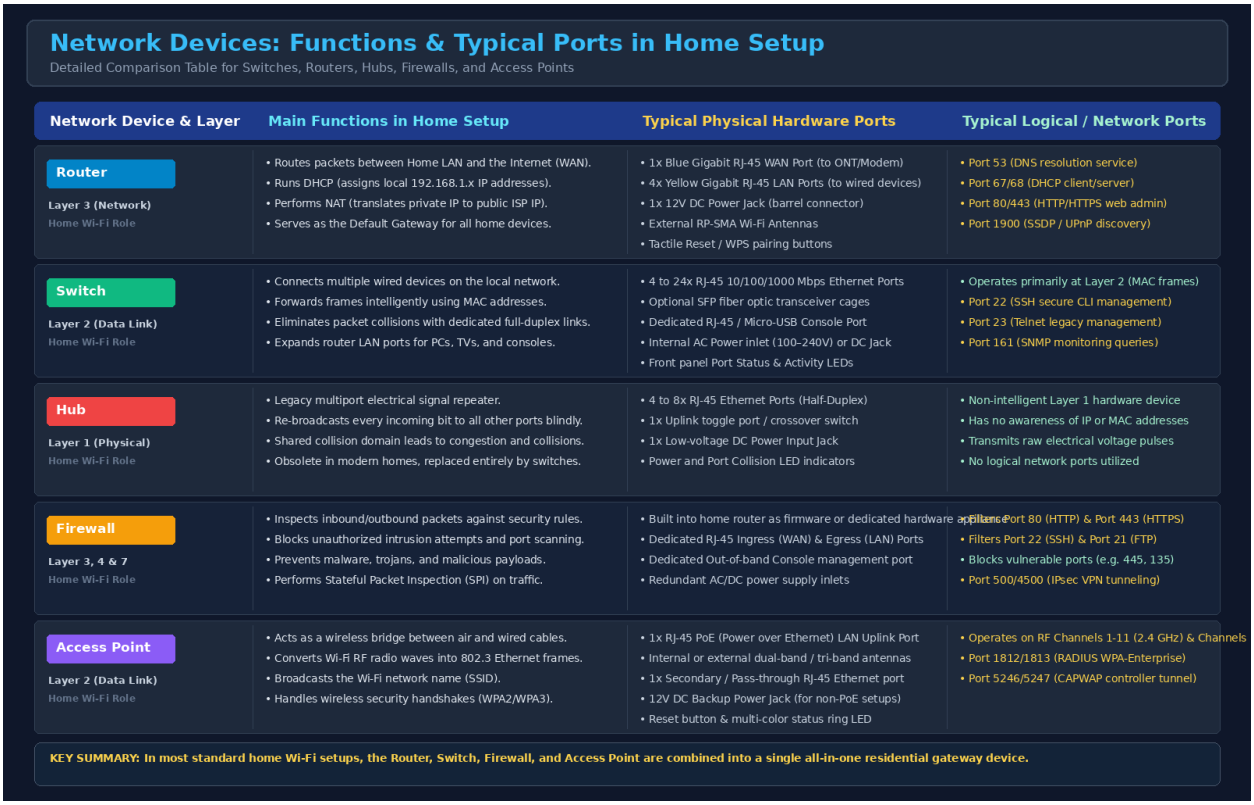
Introduction to Network Devices

Task 1: Comparison Table of Network Devices, Functions, and Typical Ports



Network Devices: Functions & Typical Ports in Home Setup

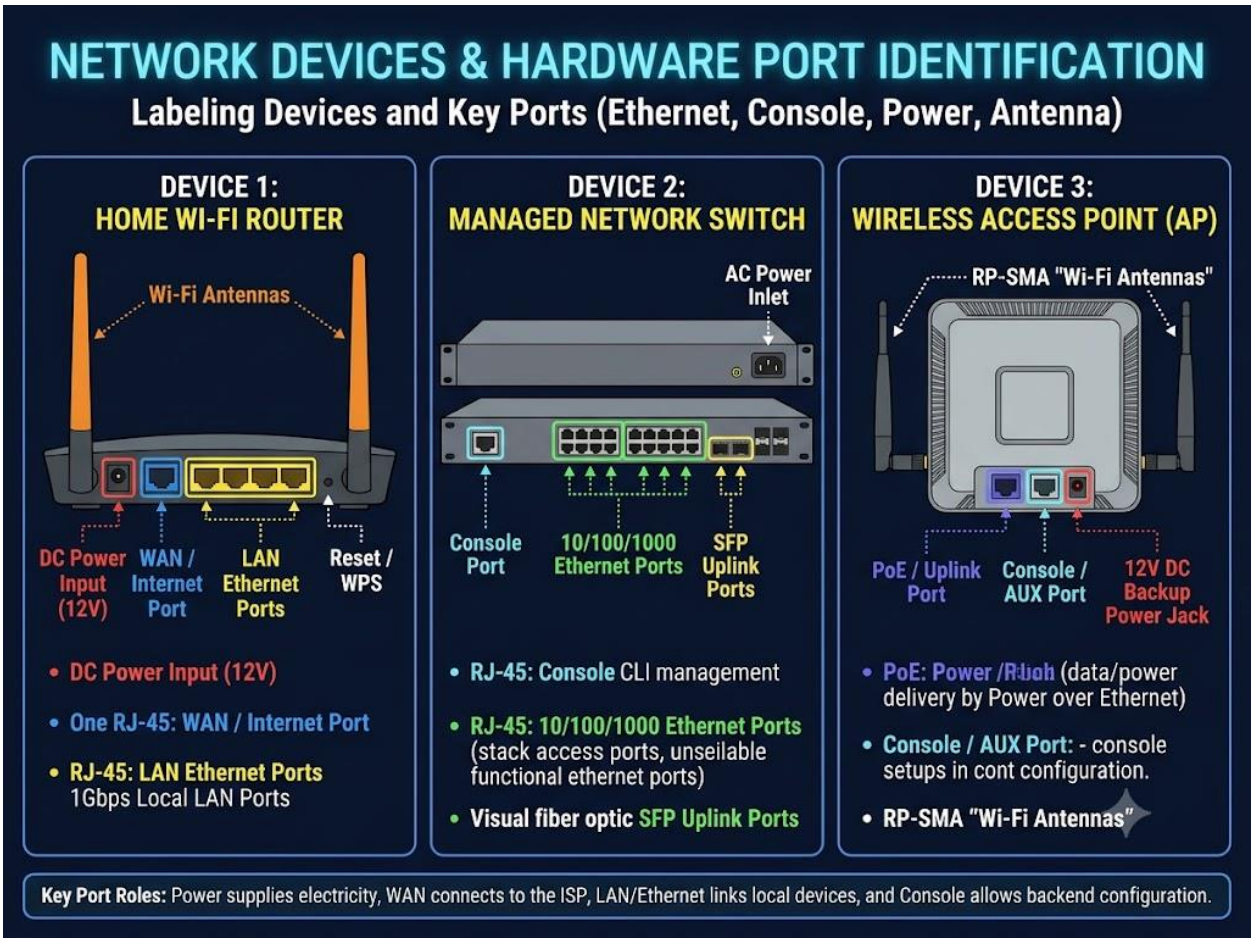
Detailed Comparison Table for Switches, Routers, Hubs, Firewalls, and Access Points

Network Device & Layer	Main Functions in Home Setup	Typical Physical Hardware Ports	Typical Logical / Network Ports
Router Layer 3 (Network) Home Wi-Fi Role	• Routes packets between Home LAN and the Internet (WAN). • Runs DHCP (assigns local 192.168.1.x IP addresses). • Performs NAT (translates private IP to public ISP IP). • Serves as the Default Gateway for all home devices.	• 1x Blue Gigabit RJ-45 WAN Port (to ONT/Modem) • 4x Yellow Gigabit RJ-45 LAN Ports (to wired devices) • 1x 12V DC Power Jack (barrel connector) • External RP-SMA Wi-Fi Antennas • Tactile Reset / WPS pairing buttons	• Port 53 (DNS resolution service) • Port 67/68 (DHCP client/server) • Port 80/443 (HTTP/HTTPS web admin) • Port 1900 (SSDP / UPnP discovery)
Switch Layer 2 (Data Link) Home Wi-Fi Role	• Connects multiple wired devices on the local network. • Forwards frames intelligently using MAC addresses. • Eliminates packet collisions with dedicated full-duplex links. • Expands router LAN ports for PCs, TVs, and consoles.	• 4 to 24x RJ-45 10/100/1000 Mbps Ethernet Ports • Optional SFP fiber optic transceiver cages • Dedicated RJ-45 / Micro-USB Console Port • Internal AC Power inlet (100–240V) or DC Jack • Front panel Port Status & Activity LEDs	• Operates primarily at Layer 2 (MAC frames) • Port 22 (SSH secure CLI management) • Port 23 (Telnet legacy management) • Port 161 (SNMP monitoring queries)
Hub Layer 1 (Physical) Home Wi-Fi Role	• Legacy multiport electrical signal repeater. • Re-broadcasts every incoming bit to all other ports blindly. • Shared collision domain leads to congestion and collisions. • Obsolete in modern homes, replaced entirely by switches.	• 4 to 8x RJ-45 Ethernet Ports (Half-Duplex) • 1x Uplink toggle port / crossover switch • 1x Low-voltage DC Power Input Jack • Power and Port Collision LED indicators	• Non-intelligent Layer 1 hardware device • Has no awareness of IP or MAC addresses • Transmits raw electrical voltage pulses • No logical network ports utilized
Firewall Layer 3, 4 & 7 Home Wi-Fi Role	• Inspects inbound/outbound packets against security rules. • Blocks unauthorized intrusion attempts and port scanning. • Prevents malware, trojans, and malicious payloads. • Performs Stateful Packet Inspection (SPI) on traffic.	• Built into home router as firmware or dedicated hardware appliance • Dedicated RJ-45 Ingress (WAN) & Egress (LAN) Ports • Dedicated Out-of-band Console management port • Redundant AC/DC power supply inlets	• Filters Port 80 (HTTP) & Port 443 (HTTPS) • Filters Port 22 (SSH) & Port 21 (FTP) • Blocks vulnerable ports (e.g. 445, 135) • Port 500/4500 (IPsec VPN tunneling)
Access Point Layer 2 (Data Link) Home Wi-Fi Role	• Acts as a wireless bridge between air and wired cables. • Converts Wi-Fi RF radio waves into 802.3 Ethernet frames. • Broadcasts the Wi-Fi network name (SSID). • Handles wireless security handshakes (WPA2/WPA3).	• 1x RJ-45 PoE (Power over Ethernet) LAN Uplink Port • Internal or external dual-band / tri-band antennas • 1x Secondary / Pass-through RJ-45 Ethernet port • 12V DC Backup Power Jack (for non-PoE setups) • Reset button & multi-color status ring LED	• Operates on RF Channels 1-11 (2.4 GHz) & Channels • Port 1812/1813 (RADIUS WPA-Enterprise) • Port 5246/5247 (CAPWAP controller tunnel)

KEY SUMMARY: In most standard home Wi-Fi setups, the Router, Switch, Firewall, and Access Point are combined into a single all-in-one residential gateway device.

Task 2: Network Devices with Labeled Hardware Ports

The diagram below illustrates the physical front/rear panels and labeled ports for three network devices:



Labeled Hardware Port Breakdown:

1. **Home Wi-Fi Router:**
    - **Antennas:** External dual-band omnidirectional antennas broadcasting 2.4 GHz2.4\text{ GHz}2.4 GHz and 5 GHz5\text{ GHz}5 GHz radio waves.
    - **Blue WAN Port (RJ-45 Ethernet):** Connects to the ISP optical modem (ONT) to receive public internet connectivity.
    - **Yellow LAN Ports (RJ-45 Ethernet):** Switched local ports to plug in desktop computers, smart TVs, or switches.
    - **Power Port:** 12V DC barrel jack connecting to the AC/DC wall power adapter.

- **Reset/WPS Button:** Tactile button to trigger quick wireless pairing or reset device firmware.

2. **Managed Network Switch:**
    - **FastEthernet / Gigabit Ports (RJ-45):** 10/100/1000 Mbps access ports for end devices, configured into VLANs.
    - **Console Port (RJ-45):** Dedicated serial port for out-of-band CLI management via a rollover cable connected to an engineer's laptop running PuTTY.
    - **Power Socket:** Standard IEC three-prong grounded AC inlet (100–240V).
3. **Wireless Access Point (AP):**
    - **PoE Uplink Port (RJ-45 Ethernet):** Power-over-Ethernet port that receives both data and electric operating current over a single Cat6 cable from a PoE switch.
    - **RF Antennas:** Internal or external antenna arrays transmitting Wi-Fi 6 (802.11ax802.11\text{ax}802.11ax) coverage.
    - **Console / Aux Port:** Serial interface for initial configuration or troubleshooting.
    - **12V DC Jack:** Backup power socket used when the upstream switch does not supply PoE.

Task 3: Real-World App Role: Ordering Food on Zomato

When you place an order on **Zomato** from your smartphone:

- **Wireless Access Point (AP):** Captures the encrypted Wi-Fi radio signals transmitted from your phone's antenna and translates them into wired Ethernet frames without signal loss.
- **Firewall:** Inspects the outbound order request to confirm it is legitimate HTTPS web traffic (Port 443), while blocking unauthorized incoming connection attempts and malicious payloads from reaching your phone.
- **Router:** Directs the packet from your private home subnet (192.168.1.15) across your ISP's network to Zomato's AWS cloud server using Public IP routing, and directs the order confirmation receipt back to your phone.

Task 4: College Wi-Fi Troubleshooting Scenario

1. The Core Problem (Bottleneck Identification):

- In a college campus, hundreds of students gather in classrooms, cafeterias, and libraries, each carrying a phone, laptop, and smartwatch.
- **Root Cause — Access Point Client Over-Capacity & Co-Channel Interference:**

Most consumer or entry-level Access Points (APs) can only reliably manage **30 to 50 active client associations** at a time. When 150+ students connect to a single AP, the shared radio frequency airtime becomes oversaturated, leading to severe packet collisions, high latency, dropped connections, and automatic disconnections.

2. The Solution (Device Upgrade):

- **Upgrade the Wireless Access Point to an Enterprise Wi-Fi 6/6E Multi-Radio AP (e.g., Cisco Catalyst or Aruba AP):**
    - **OFDMA & MU-MIMO:** Wi-Fi 6 allows the AP to communicate with up to dozens of client devices concurrently within a single transmission cycle, eliminating the queue delay where devices take turns.
    - **Higher Device Density:** Enterprise APs are engineered with dedicated co-processors and increased RAM to handle **250+ simultaneous active devices** without crashing or dropping connection state tables.
    - **BSS Coloring:** Mitigates channel overlap and co-channel interference from neighboring APs across nearby lecture halls.